

Bharath Kesav R

+91 7418297053 | bharathkesav1275@gmail.com

[in LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [HuggingFace](#)
Pollachi, Coimbatore, Tamil Nadu – 642114

OBJECTIVE

Integrated M.Sc. Data Science student at Amrita Vishwa Vidyapeetham with hands-on experience in data analytics, machine learning, deep learning, and LLM-based AI systems. Built 5+ research-grade end-to-end pipelines spanning Graph Attention Networks, RAG + LLaMA 3.1, multi-task Transformers, IoT sensor analytics, and SHAP-driven explainability processing datasets from millions of IoT readings to 25,000+ football match records. Proficient in Python, PyTorch, scikit-learn, HuggingFace Transformers, SHAP, SQL, Power BI, and Streamlit, backed by 10 public GitHub repositories and 5+ live HuggingFace deployments. Seeking an internship in data analytics, data science, or AI/ML engineering to deliver scalable, explainable, and insight-driven solutions to real-world problems.

EDUCATION

• Amrita Vishwa Vidyapeetham

Int. M.Sc. Data Science

Sept 2022 – June 2027

Coimbatore, Tamil Nadu

- **Relevant Coursework:** Business Analytics, Statistical Quality Control, Database Management Systems, Big Data Analytics, Data Wrangling, Machine Learning, Deep Learning, NLP, Probability & Statistics, Linear Algebra

PROJECTS

• Explainable Match Summaries — RAG + LLM + SHAP Reporting Pipeline

Tools: Sentence-BERT, FAISS, Groq API, LLaMA 3.1, SHAP, Pandas, Streamlit

Jan 2026 – Apr 2026

[\[🔗\]](#) [\[Live Demo\]](#)

- Independently curated a structured dataset of 189 UCL-2025 matches × 142 columns from primary sources — achieving mean retrieval cosine similarity **0.903 vs 0.373 baseline** with FAISS-powered semantic search
- Deployed LLaMA 3.1 via Groq API for hallucination-free grounded summary generation; SHAP LinearExplainer identified goal difference as the strongest match outcome predictor (mean |SHAP| = 1.000)

• Sarcasm Detection via Sentiment Incongruence — Multi-Task NLP

Tools: HuggingFace Transformers, PyTorch, LIME, scikit-learn, Pandas

Feb 2026 – Apr 2026

[\[🔗\]](#) [\[Live Demo\]](#)

- Proposed a novel **Sentiment Incongruence Auto-Labeler** deriving dataset-agnostic sarcasm labels from semantic mismatch between surface sentiment and underlying emotion using two parallel Transformer pipelines
- Fine-tuned roberta-base with joint Focal Loss and WeightedRandomSampler achieving **F1-Macro = 0.977** and **AUC = 0.997** on 4,293 test samples; LIME token-level rationale extraction accompanies every prediction

• Football Match Outcome Prediction — BI Dashboard Across 5 Leagues

Tools: Python, scikit-learn, SQL, SHAP, Pandas, Streamlit

Jan 2026 – Mar 2026

[\[🔗\]](#) [\[Live Demo\]](#)

- Engineered a structured data pipeline processing **25,979 historical records** across 5 European leagues — computing 33 statistical KPIs including ELO ratings, rolling form metrics, and normalised betting probabilities with a three-layer anti-leakage architecture
- Deployed an interactive Streamlit dashboard covering 259 teams; Gradient Boosting outperformed ELO baseline by +0.073 F1-Macro; SHAP explainability translated model outputs into clear, business-readable analytical reports

• Tyre Manufacturing Quality Analytics Dashboard

Tools: Python, Pandas, Matplotlib, Seaborn, scikit-learn, SHAP, Streamlit

May 2026 – Jun 2026

[\[🔗\]](#) [\[Live Demo\]](#)

- Built an end-to-end manufacturing audit analytics dashboard on 3,240 production batches — computed 10 KPIs including OEE (85.37%), cost per unit, and defect rate; applied X̄-R control charts with $\pm 3\sigma$ limits identifying out-of-control months for audit escalation
- Deployed a **Batch Risk Scorer** (Gradient Boosting + SHAP, ROC-AUC 0.7417) enabling auditors to input operational parameters and receive an instant risk classification (Low/Medium/High) with feature-level explainability

SKILLS

- **Languages & Data:** Python, SQL, LaTeX, HTML, CSS; Pandas, NumPy, Matplotlib, Seaborn, Plotly; Data Cleaning, Feature Engineering, Data Pipeline Development
- **Analytics & Visualization:** Power BI, Tableau, MS Excel, KPI Reporting, EDA Dashboards, Statistical Analysis, Hypothesis Testing, SPC Control Charts, SHAP Explainability, Data Storytelling
- **Machine Learning:** scikit-learn, Gradient Boosting, Random Forest, Logistic Regression, SVM, SMOTE, Cross-Validation, ELO Ratings, TimeSeriesSplit
- **Deep Learning & NLP:** PyTorch, TensorFlow, HuggingFace Transformers, RoBERTa, BiLSTM, GAT, Multi-task Learning, Focal Loss, LIME, Attention Visualization
- **LLMs & Retrieval:** LLaMA 3.1, RAG, FAISS, Sentence-BERT, Groq API, Prompt Engineering, Streamlit
- **Tools & Platforms:** Git, GitHub, PostgreSQL, SQLite, Kaggle GPU, Google Colab, Jupyter Notebook, VS Code, Overleaf

ACHIEVEMENTS

• Finalist — Pegasus 3.0, IEEE Student Branch Technical Competition

Muthoot Institute of Technology and Science (MITS)

Jan 2025

[\[🔗\]](#)

- Selected as Finalist at Pegasus 3.0 for analytical problem framing and technical execution; developed an NLP-powered Study Planner for students with cognitive shortcomings (ADHD, dyslexia) enabling personalised task management

ADDITIONAL INFORMATION

Languages: Tamil (Native), English (Professional), Malayalam (Basic)
LLMs, Open-Source Development

Interests: Sports Analytics, Data Analytics, AI/ML Research,